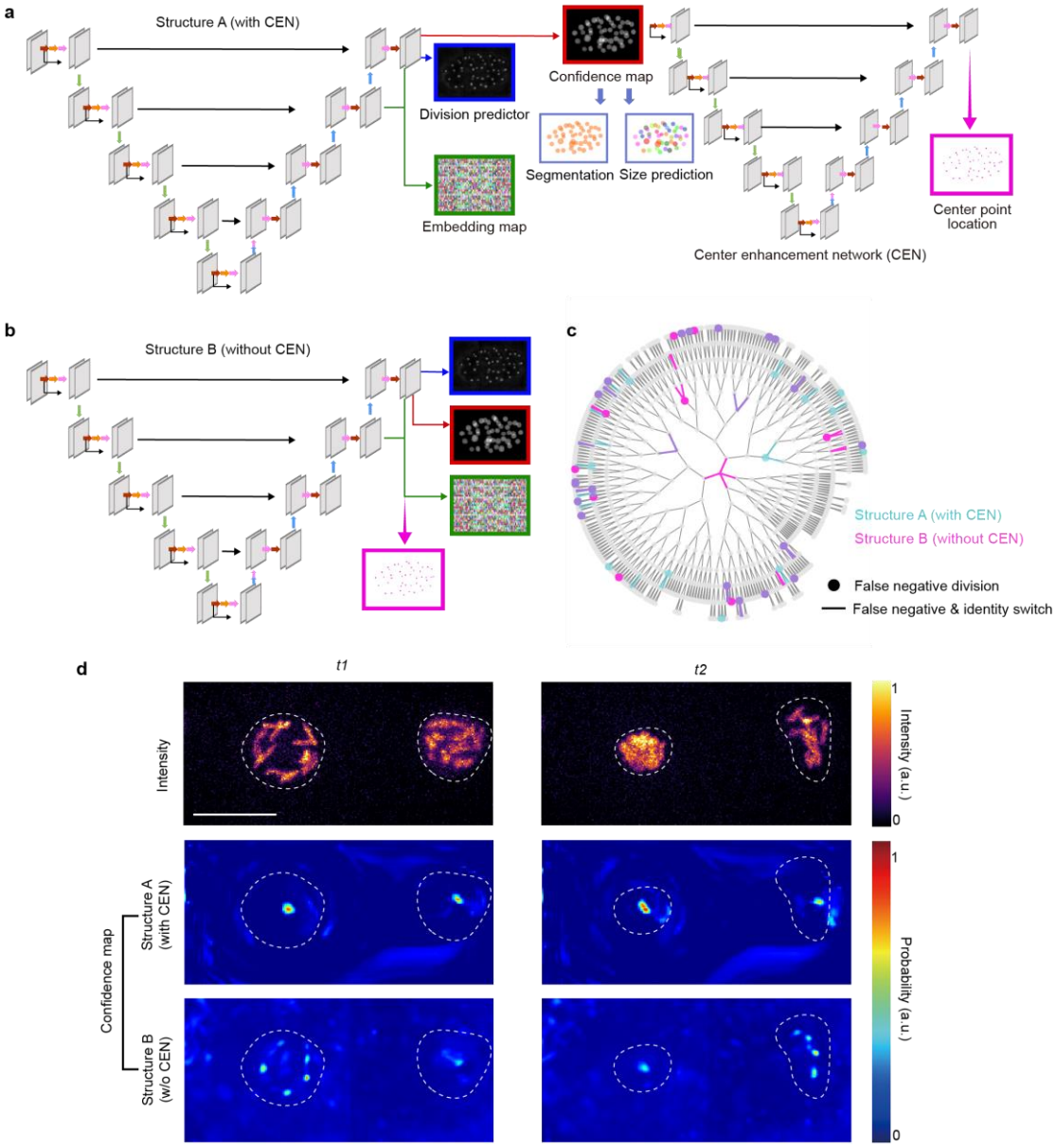


Supplementary Figure 7 | Ablation study of CEN module on tracking performance. a, The CELLECT model structure used in this study. **b**, A modified version of the CELLECT model without the CEN subnetwork. While the CEN structure was removed, its training methodology and output format were retained to mitigate performance drop. Without these modifications, simply removing the CEN would result in outcomes similar to the 2-level configuration output scenario, leading to significantly worse performance and making it difficult to compare model structures. **c**, We validated the performance differences of 2 methods on the mskcc-confocal dataset[1,2] #2. As shown, the model without the CEN module produces more errors, with incorrect links appearing even in the early stages. **d**, A comparison of the confidence maps of the 2 models in the early stages. We found that the CEN module shows more accurate performance of center extraction especially for larger cells, thus more efficiently reduces incorrect cell linking. All evaluation data in this figure were drawn from mskcc-confocal dataset #2. All models were trained using mskcc-confocal dataset #3. Scale bars: 10 μm.